SREEPERUMBUDURU AKSHIT

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EDUCATION

B V Raju Institute of Technology, JNTU Hyderabad **Hyderabad, India**
Bachelor of Technology (b.tech) in computer science and engineering *2023-2027*
Cumulative GPA – 8.22/10.0 *(pursuing)*

Excellencia Junior College Telangana State Board of Intermediate Education (TSBIE) **Hyderabad, India**
Intermediate –93.1% *2021-2023*

PROJECTS

Project: Smart Wearable System for Early Internal Bleeding Detection

Description: Developed a Flutter-based mobile app, VitalGuardians, to monitor key vital signs such as hemoglobin levels, blood oxygen, heart rate, and blood pressure.

Skills/Tools: Flutter, Dart, mobile app development, wearable sensor, data integration.

- Developed a Flutter-based mobile app, **VitalGuardians**, to monitor key vital signs: Hemoglobin levels, Blood oxygen saturation (SpO_2), Heart rate Blood pressure, Bioimpedance, Pulse.
- Calculated a bleeding risk score and classified risk levels (low, moderate, high) using rule-based algorithms for early detection of internal bleeding.
- Provided non-invasive, real-time alerts to users without the need for implants.
- Designed the system with plans for future AI and machine learning integration to improve predictive accuracy.
- Tools & Technologies: Flutter, Dart, mobile app development, basic algorithm design, wearable sensor data integration.
- Project showcased at a hackathon; video demonstration available here: LinkedIn Video by Akshit Sreeperumbuduru

Project: Detecting Deepfake Images using Deep Learning and Generative Adversarial Networks

Description: Developed a hybrid deep learning framework to classify images as *Real* or *Deepfake*, combining feature-based classification with generative modeling. Published in **IEEE** at the **5th International Conference on Soft Computing for Security Applications (ICSCSA-2025)**.

Skills/Tools: XceptionNet (CNN), Support Vector Machine (SVM), Autoencoder GAN (AGAN), Python, TensorFlow/Keras, Scikit-learn, Computer Vision, Machine Learning.

- Implemented **feature extraction** using pretrained **XceptionNet (CNN)**.
- Applied **SVM classifier** for binary classification (Real vs. Fake images).
- Designed an **Autoencoder GAN** trained on real images for anomaly detection.
- Built a **hybrid fusion system** combining SVM probabilities with GAN reconstruction error for improved robustness.
- Achieved **~99% validation accuracy** and **~93% test accuracy** on unseen test images.
- Processed and augmented a Kaggle dataset of **2,100+ real and fake images**.
- Addressed the growing challenge of **deepfake detection** for AI security and trustworthiness.
- Project post: LinkedIn Post by Akshit Sreeperumbuduru

Project: Hospital AI Chatbot for Patient Data Analysis

Description: Developed a web-based AI chatbot that analyzes patient datasets and acts as a medical assistant by answering healthcare-related queries.

Skills/Tools: Python, Flask, Pandas, Matplotlib, HuggingFace API, NLP

- Built an interactive chatbot to query patient datasets using natural language
- Implemented data validation (missing values, duplicates, data types) before processing

- Enabled dynamic question answering over structured healthcare data
- Integrated visualization (bar charts) for vitals like heart rate and blood pressure
- Developed backend using Flask and connected it with a simple web interface
- Improved accessibility of healthcare data by converting raw datasets into actionable insights
- Project post: LinkedIn Post by Akshit Sreeperumbuduru

SKILLS

- **Programming Languages:** Python (intermediate), C (intermediate), Java (Beginner) , SQL (Beginner)
- **Software Tools/IDEs:** Jupyter Notebook and VS-Code
- **Microsoft Office Skills:** Word, Excel and PowerPoint
- **Management & Leadership Skills:** Mentoring, Brainstorming, Mind Mapping and Leading
- **Other skills:** Data Structures and Algorithms, Machine Learning, Object-Oriented Programming, Database Management Systems

ACHIEVEMENTS

- **AI/ML in Geospatial Data Analysis**
  Indian Institute of Remote Sensing (IIRS)
  View Certificate
- **Supervised Machine Learning: Regression and Classification**
  Coursera (offered by Stanford University)
  Verify Certificate
- **Introduction to Artificial Intelligence**
  Infosys Springboard
  View Certificate

EXTRA-CURRICULAR ACTIVITIES

PROMETHEAN 2025 Hackathon

- Collaborated with a diverse team to develop innovative technology solutions within PROMETHEAN 2025, a competitive hackathon focused on teamwork and rapid problem-solving.
- Demonstrated skills in coding, creativity, and communication to build functional prototypes under tight deadlines.
- Gained hands-on experience in product development and collaborative innovation.
- LinkedIn Post by Akshit Sreeperumbuduru highlighting teamwork and innovation during the event.

HACKFINITI 2025 National-Level Hackathon

- Developed the BleedSense project focused on early detection of internal bleeding using wearable technology.
- Collaborated with a team in a 24-hour intensive hackathon organized by B V Raju Institute of Technology and Infiniti Tech.
- Applied problem-solving, rapid prototyping, and teamwork skills to design an innovative health monitoring solution.
- Gained valuable experience in real-time product development, innovation, and peer collaboration.